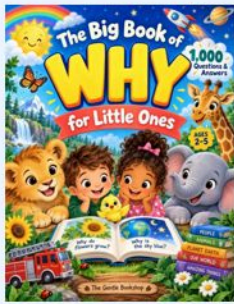


The Curious Kids Club

Free Sampler

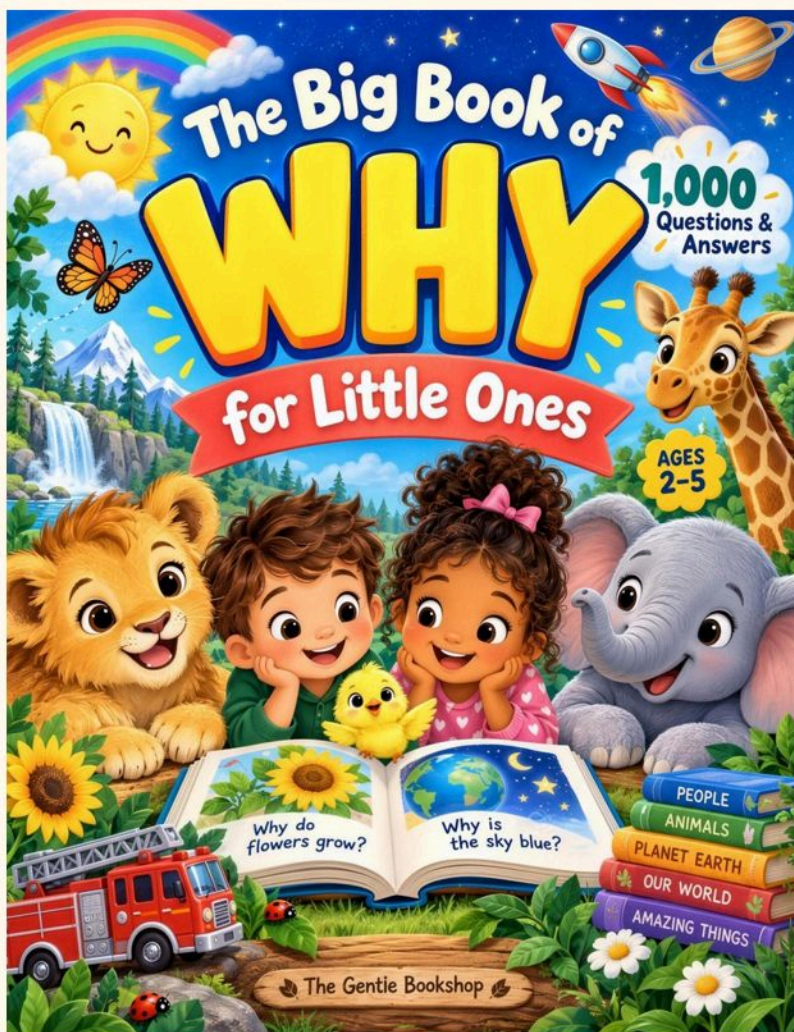
25 pages from 5 of our fact books:
why questions, gross facts and what-if adventures
for curious kids aged 2 to 12.



Scan to play games and hear answers read aloud
Free for home and classroom use. Please do not sell it.

Big Book of Why for Little Ones

Ages 2 to 5



5 sample pages follow. Tap this page to see the whole book.

MY AMAZING BODY



Why do I have fingernails?

Nails protect the ends of your fingers and help you pick up tiny things. They grow all the time.



Why does my tummy rumble?

Your tummy squeezes to move food along, and when it is empty the squeezes make a growly noise. It means 'feed me!'.



Why do I burp?

When you eat or drink you swallow some air. A burp is that air coming back up, so it does not sit in your tummy.



Why do I get hot when I run?

Your muscles make heat when they work hard, like a warm engine. Sweat helps cool you down again.



Why do I sweat?

Sweat is water your skin lets out when you are hot. As it dries it takes the heat away with it.



Why do I have a tongue?

Your tongue tastes food, moves it around while you chew, and helps you talk. Try saying "lion" without it!

PETS AND FARM



Why do cats sleep so much?

Cats are hunters, and hunting takes lots of energy. Long naps save it up for pouncing.



Why do cats land on their feet?

A cat can twist its bendy body in the air super fast, so its feet point down before it lands.



Why do rabbits have long ears?

Long ears hear danger from far away, and they let out heat to keep the rabbit cool.



Why do rabbits twitch their noses?

Twitching pulls in more smells. A rabbit's nose is always checking for food and foxes.



Why do hamsters stuff their cheeks?

Hamsters have big stretchy pockets in their cheeks to carry food back to their nest.



Why do goldfish open and close their mouths?

Fish breathe by gulping water and pushing it over their gills. That is how they get air from water.

WILD ANIMALS



Why do baby zebras stay close to mum?

A newborn foal learns its mother's own stripe pattern within days, so it never loses her in a big herd. Staying close also keeps it safe from lions.



Why do monkeys live in trees?

Trees keep monkeys safe from animals that cannot climb. Branches also give them fruit, leaves and a comfy place to nap.



Why do monkeys have tails?

Many monkeys use their tail like a spare hand for gripping branches. A good grip means they will not fall while climbing high.



Why do monkeys groom each other?

Picking through fur removes bugs and dirt, and it feels nice too. Grooming is also how monkey friends and families show they care.



Why do monkeys chatter and screech?

Loud calls warn the whole troop when danger like a snake or eagle is near. Different calls can mean different kinds of danger.



Why do gorillas beat their chests?

A chest beat is how a gorilla shows he is strong or a little worried. It is mostly for show, not for fighting.

SPACE



Why is the sun shaped like a ball?

All that hot glowing gas pulls evenly inward from every side. Pulling equally from all directions squeezes the sun into a ball.



Why does the sun look yellow?

Sunlight is really every colour mixed together into one bright glow. Travelling through our air makes that light look yellow to us.



Why is the sun so enormous?

The sun is so huge that over a million Earths could fit inside it. Next to the sun, our whole planet is just a tiny speck.



Why is the sun our closest star?

Of all the stars twinkling at night, the sun is nearest to us. That is why it looks so much bigger and brighter than the rest.



Why does the moon come out at night?

As our side of Earth turns away from the sun it grows dark. The moon is up there too, just easier to spot after dark.



Why does the moon shine at night?

The moon has no light of its own at all. It simply bounces sunlight back down to us like a mirror.

PLANTS AND THE GARDEN



Why do some trees stay green all winter?

Trees called evergreens have thin, tough, waxy needles that can survive frost without falling off. That way they keep making food from sunlight even in winter.



Why do trees have so many branches?

Spreading branches hold up lots of leaves in different spots, so the tree can catch more sunlight all round. More sunlight means more food for the tree.



Why are leaves green?

Leaves are packed with a green colouring that traps sunlight and turns it into food for the plant. That busy green stuff is what gives leaves their colour.



Why are most leaves flat and wide?

A flat, wide shape gives a leaf plenty of open surface to soak up as much sunlight as it can. More sunlight caught means more food made.



Why do leaves have lines running through them?

Those little lines are veins that carry water up into the leaf and carry food back out to the rest of the plant.

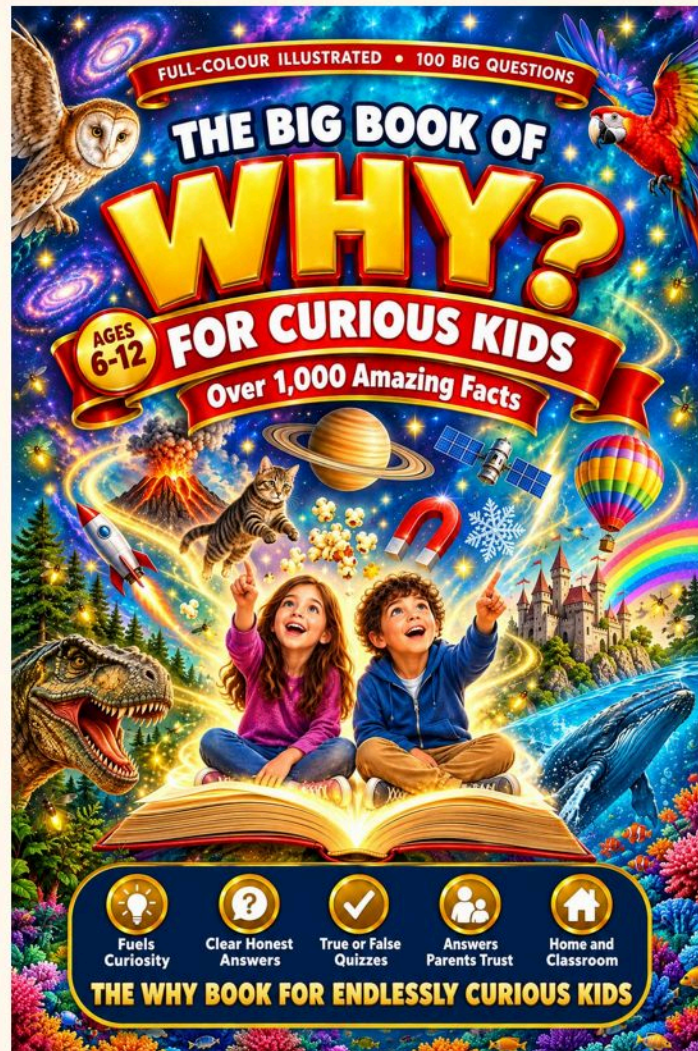


Why do plants need leaves at all?

Leaves are where a plant turns sunlight, air and water into the food it needs to grow big and strong.

Big Book of Why for Curious Kids

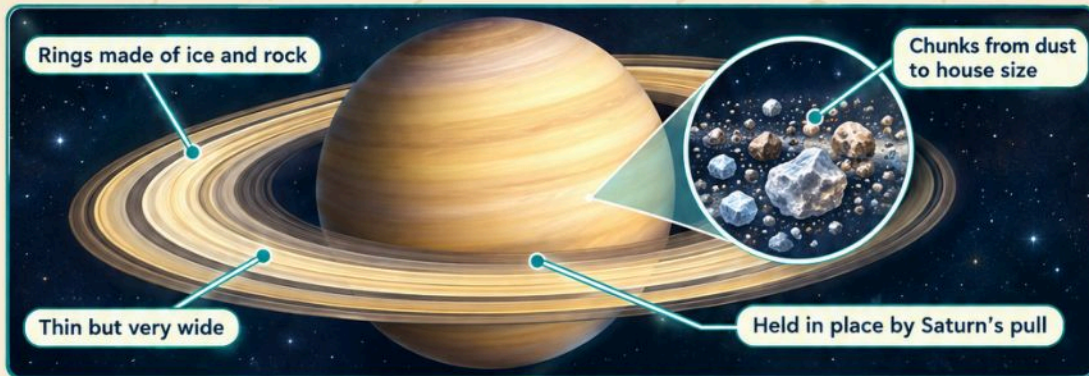
Ages 6 to 12, full colour



5 sample pages follow. Tap this page to see the whole book.

SPACE

Why does Saturn have rings?



WOW FACT

If Saturn were the size of a football, its rings would be thinner than a sheet of paper.

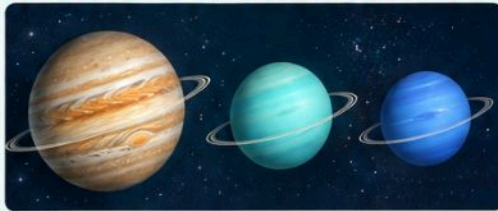
What's going on?

Saturn's rings are not solid. They are made of countless pieces of ice and rock, some as small as dust and some as big as a house, all circling the planet at high speed. Saturn's strong gravity keeps them in place and spreads them out into flat, wide bands. Scientists think the rings are the remains of a moon or comet that came too close and was pulled apart, or leftover material that never joined together to make a moon. They are still working out which.

In a nutshell

- ✓ The rings are billions of separate ice and rock pieces.
- ✓ Saturn's gravity flattens them into wide, thin bands.
- ✓ All four giant planets have rings of some kind.

Not the only ringed planet



Jupiter, Uranus and Neptune have rings too. They are dark and faint, so they are hard to see.

A moon torn apart



A moon that drifts too close can be pulled to pieces. The pieces spread out and become ring material.

True or false?



Jupiter has rings too.

TRUE.

Jupiter, Uranus and Neptune all have faint rings. Saturn's are just the brightest.



The rings are one solid disc.

FALSE.

They are loose pieces, each one going round on its own.

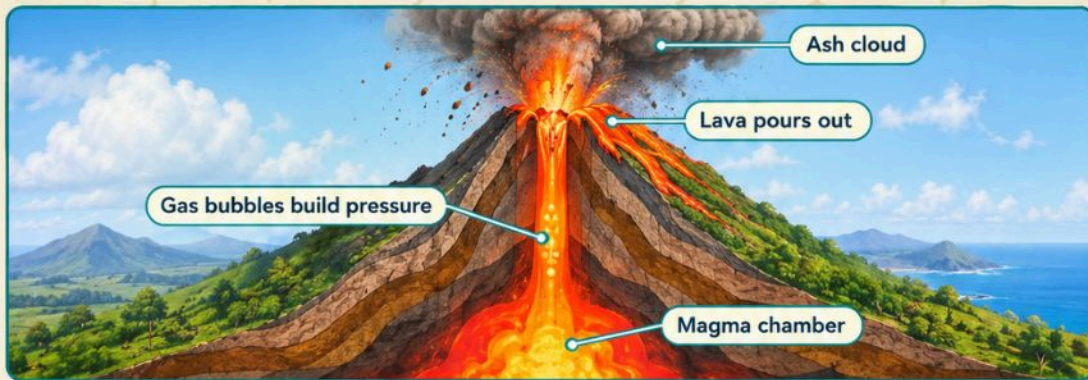


Scientists know exactly how the rings formed.

FALSE.

They have good ideas but are still not sure.

Why do volcanoes erupt?



WOW FACT

There are about 1,500 volcanoes on land that could erupt, and many more hidden under the sea.

What's going on?

Deep under the ground the rock is so hot that in places it melts. Melted rock is called magma, and it is lighter than the solid rock around it, so it slowly rises and collects in a chamber under the volcano. Magma is full of gas, like a fizzy drink in a bottle. As more magma pushes up, the pressure grows and grows. When the pressure is too much for the rock above to hold, the volcano bursts open and the magma shoots out. Once it is outside, it is called lava.

In a nutshell

- ✓ Hot melted rock rises and collects under the volcano.
- ✓ Gas in the magma builds up pressure.
- ✓ The pressure blasts through the top.

Runny or sticky



Runny lava oozes out gently and builds wide volcanoes. Sticky lava traps gas and explodes.

New land



Cooled lava becomes new rock. Whole islands, like Iceland and Hawaii, were built by volcanoes.

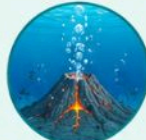
True or false?



Lava is rock that is burning.

FALSE.

Lava is rock that has melted. It glows because it is so hot, not because it is on fire.



Most volcanoes are under the sea.

TRUE.

The sea floor has far more volcanoes than the land does.



A quiet volcano can never erupt again.

FALSE.

Some volcanoes sleep for hundreds of years and then wake up.

Why is every snowflake different?



WOW FACT A single snowflake can contain about a billion billion water particles.

What's going on?

A snowflake begins as a tiny speck of dust high in a cloud. Water freezes onto it, and because of the way water particles lock together, they always build a six-sided shape. As the flake tumbles down through the cloud, it passes through pockets of air that are slightly warmer, colder, wetter or drier, and each change makes it grow a little differently. No two flakes take exactly the same path through the cloud, so no two grow exactly the same arms. Every flake is six-sided, but every flake is its own design.

In a nutshell

- ✓ Water always freezes into six-sided shapes.
- ✓ Each flake grows as it falls through changing air.
- ✓ No two flakes take the same path, so no two match.

Snowflake shapes



Plates, stars, needles and columns all form depending on how cold and how wet the cloud is.

The snowflake photographer



In 1885 a farmer called Wilson Bentley took the first photographs of snowflakes, over 5,000 of them in his life.

True or false?



Snowflakes can have any number of points.

FALSE.

Real snowflakes have six sides, because of how water freezes.



Every snowflake starts on a speck of dust.

TRUE.

Water needs something to freeze onto. Dust or pollen does the job.



Snow is white because the ice is white.

FALSE.

Ice is clear. Snow looks white because all the tiny crystals scatter light in every direction.

Why does popcorn pop?


WOW FACT

A popcorn kernel can jump almost a metre into the air when it pops.

What's going on?

Every popcorn kernel has a hard, waterproof shell with soft starch and a tiny drop of water sealed inside. When you heat it, the water turns into steam, but the steam is trapped, so the pressure inside builds and builds. At about 180 degrees the shell can no longer hold on and it bursts open. The steam escapes in a flash, the soft starch inside puffs out and sets into a fluffy white cloud, and the kernel flips inside out. That sudden burst is the pop you hear.

In a nutshell

- ✓ Each kernel has water sealed inside a hard shell.
- ✓ Heat turns the water to steam and pressure builds.
- ✓ The shell bursts and the starch puffs out.

Why some do not pop



A kernel with a cracked shell or too little water inside cannot build pressure, so it stays hard.

The two shapes



Some kernels pop into butterfly shapes, others into round mushrooms. Cinemas usually use the butterfly kind.

True or false?



Any corn will pop if you heat it.

FALSE.

Only popcorn has the right hard shell. Sweetcorn just dries out or burns.



Popcorn pops because of steam.

TRUE.

Trapped water turns to steam and bursts the shell.



People were eating popcorn thousands of years ago.

TRUE.

Popcorn has been found in caves in the Americas that are over 5,000 years old.

HISTORY AND LONG AGO

Why did the dinosaurs disappear?



WOW FACT

Not every dinosaur died. Birds are living dinosaurs, so there are about 10,000 kinds of dinosaur alive today.

What's going on?

About 66 million years ago a rock from space, about 10 kilometres across, smashed into the Earth near what is now Mexico. The explosion was bigger than all the world's bombs put together. It set off huge fires and giant waves, and threw so much dust and smoke into the sky that the Sun was blocked out for months. Without sunlight, plants died. Without plants, the plant-eating dinosaurs starved, and then the meat-eaters that fed on them. Small animals that could hide, dig and eat almost anything, like early mammals, survived.

In a nutshell

- ✓ A huge space rock hit the Earth 66 million years ago.
- ✓ Dust blocked the Sun and plants died.
- ✓ Big dinosaurs starved. Small creatures survived.

The crater



The crater is 180 kilometres wide and buried under rock and sea. Scientists only found it in the 1990s.

The survivors



Small mammals, birds, crocodiles and turtles came through. Mammals later grew into everything from mice to whales.

True or false?



People and dinosaurs lived at the same time.

FALSE.

The last big dinosaurs died about 65 million years before the first humans appeared.



Birds are dinosaurs.

TRUE.

They evolved from small feathered dinosaurs and never went extinct.



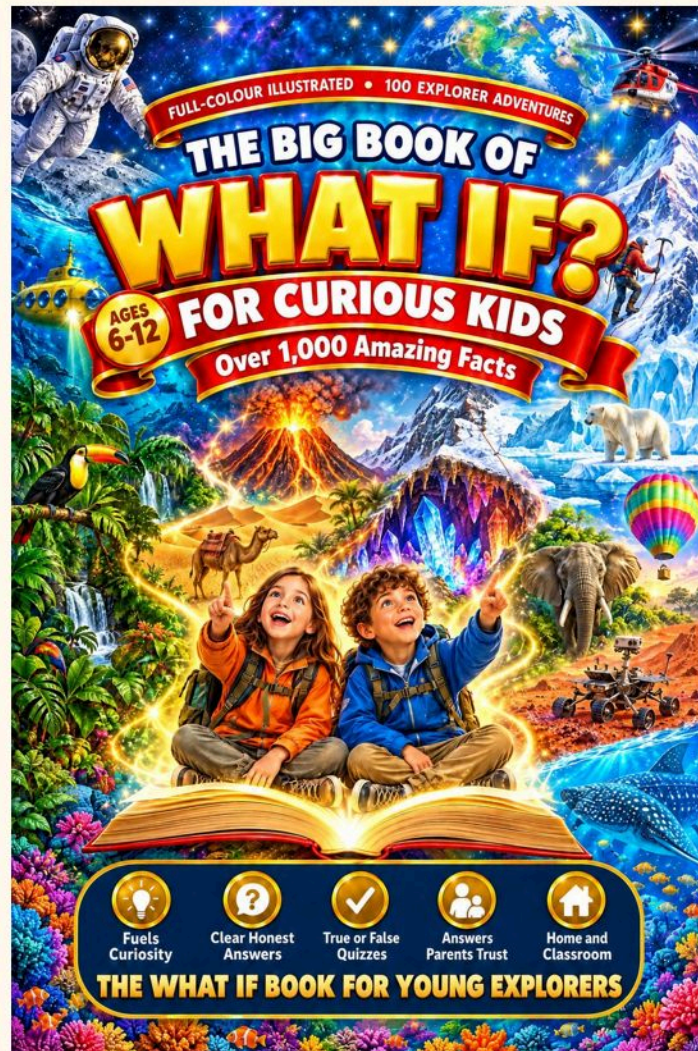
Scientists are sure an asteroid did it.

TRUE, mostly.

The crater and a worldwide layer of dust match the timing. Huge volcanoes in India may have made it worse.

The Big Book of What If?

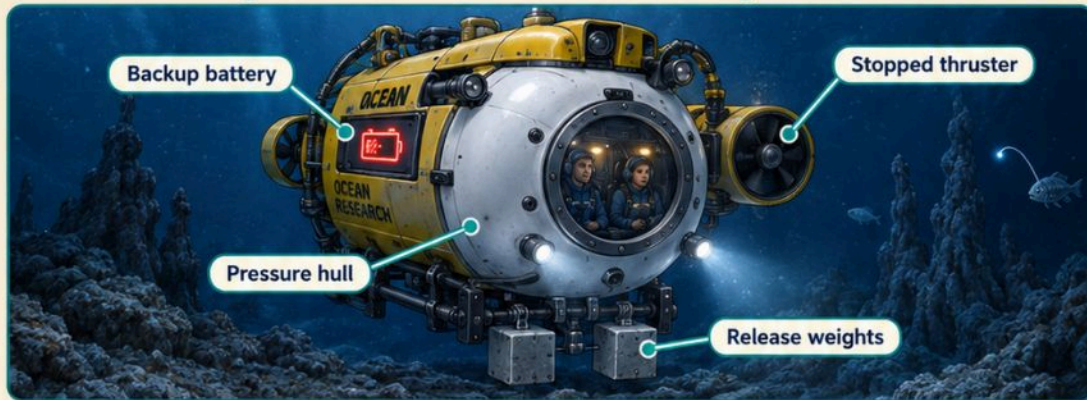
Ages 6 to 12, full colour



5 sample pages follow. Tap this page to see the whole book.

DEEP-OCEAN EXPLORERS

What if your submarine lost power in the deep ocean?



WOW FACT

Some research submersibles can release heavy weights to help them rise.

What's going on?

The lights might dim and the thrusters could stop, but losing power would not automatically crush the cabin. A strong pressure hull protects the people inside. Research submersibles carry emergency systems, and some can drop weights to become more buoyant. The crew would follow their emergency plan and try to contact the support ship. Backup air, power and careful preparation are essential.

In a nutshell

- ✓ The pressure hull protects the crew.
- ✓ Emergency systems provide backup.
- ✓ Releasing weights can help some craft rise.

A strong little room



A rounded pressure hull resists the surrounding water's squeeze.

A team above



A support ship follows the dive and helps coordinate recovery.

True or false?



No power means instant crushing.

FALSE.

Hull strength does not depend on the lights staying on.



Dropping weight can help a craft rise.

TRUE.

It can make the submersible more buoyant.



Every submarine has the same backup.

FALSE.

Emergency systems depend on the craft's design.

UNDERGROUND EXPLORERS

What if heavy rain flooded the cave you were exploring?



WOW FACT Rain falling far from a cave entrance can still send water underground.

What's going on?

A quiet cave stream can become a rushing flood when rainwater enters cracks and passages above. Narrow tunnels may fill quickly, cutting off the way out. Rain does not have to fall directly over the entrance to cause trouble. Cavers check weather and local flood risks before entering. If conditions become dangerous, trained teams follow emergency plans and call on specialist cave rescuers.

In a nutshell

- ✓ Rainwater can travel into cave passages.
- ✓ Floods can block the way out.
- ✓ Weather checks are part of planning.

Water from above



Water can move from the surface into an underground drainage system.

Planning the trip



Experienced teams check conditions and leave their trip details with someone.

True or false?



Only rain at the entrance matters.

FALSE.

Water may arrive from a wider area.



Narrow passages can fill quickly.

TRUE.

They may become dangerous barriers.



A dry entrance guarantees safety.

FALSE.

Conditions deeper inside can be very different.

POLAR EXPLORERS

What if the sea ice beneath you started breaking apart?



WOW FACT

Sea ice floats on the ocean and can move with winds and currents.

What's going on?

A solid-looking white surface can be part of a moving sheet of sea ice. Winds, currents and tides can pull it apart, opening cracks called leads. A familiar route may disappear as separate floes drift away. Polar teams assess ice conditions and plan journeys carefully. If the ice changes, trained researchers follow emergency procedures and coordinate help rather than attempting risky jumps.

In a nutshell

- ✓ Sea ice floats and moves.
- ✓ Cracks can open into wide leads.
- ✓ Changing conditions can cut off routes.

A moving surface



Separate floes can drift in different directions.

Watching the ice



Satellite images and local observations help teams assess ice conditions.

True or false?



Sea ice is fixed to the seabed everywhere.

FALSE.

Much of it floats and drifts.



Wind can move floating ice.

TRUE.

Winds and currents help drive its motion.



Yesterday's route will always remain safe.

FALSE.

Cracks and drifting floes can change the route.

DESERT EXPLORERS

What if a sandstorm swallowed your camp?



WOW FACT A dust storm can make daytime look suddenly dark.

What's going on?

Strong winds can lift sand and finer dust into the air. Visibility may fall quickly, and blowing particles can irritate eyes and breathing passages. A desert camp needs weather planning, secure equipment and suitable shelter. Trained teams follow local storm procedures and avoid unnecessary travel when they cannot see the route. Fine dust may reach places far beyond the dunes.

In a nutshell

- ✓ Wind can lift sand and fine dust.
- ✓ Poor visibility makes travel dangerous.
- ✓ Planning and shelter reduce exposure.

Different-sized particles



Fine dust can stay airborne longer than heavier sand grains.

Keeping gear protected



Careful storage helps protect essential equipment from windblown grit.

True or false?



Every airborne particle is a large sand grain.

FALSE.

Storms can carry much finer dust too.



Dust can reduce visibility.

TRUE.

Dense clouds can hide the landscape.



A clear morning guarantees a clear afternoon.

FALSE.

Weather can change during an expedition.

What if a river of glowing rock crossed your path?



WOW FACT Lava is molten rock that has reached Earth's surface.

What's going on?

Molten rock underground is called magma. Once it erupts onto the surface, it is called lava. Some lava spreads in runny streams, while thicker lava creeps into bulky mounds. Its speed depends on its properties, the slope and how it flows. Even slow lava can destroy roads and buildings. Scientists observe from planned locations because nearby heat, gases and unstable ground add danger.

In a nutshell

- ✓ Magma becomes lava at the surface.
- ✓ Different lava flows behave differently.
- ✓ Slow-moving lava can still be dangerous.

A crust on top



The surface can cool while molten rock continues moving underneath.

Watching remotely



Cameras help scientists follow changes without standing beside a flow.

True or false?



Lava and magma are unrelated materials.

FALSE.

The name changes when molten rock reaches the surface.



All lava moves at the same speed.

FALSE.

Its properties and the terrain affect its movement.



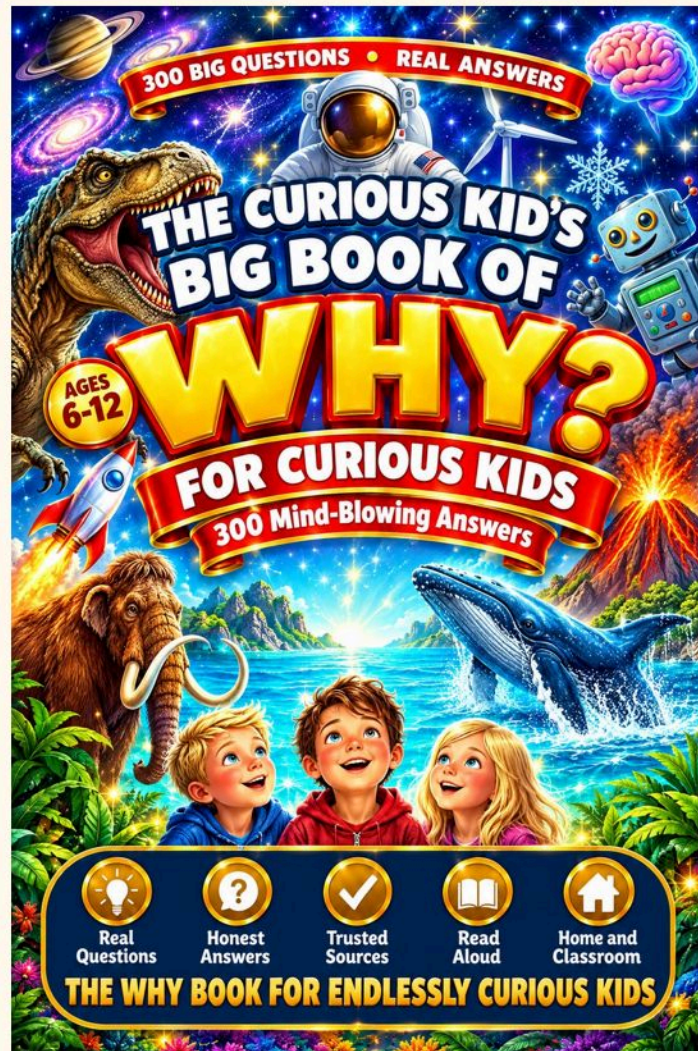
Lava cools into rock.

TRUE.

As heat escapes, molten material solidifies.

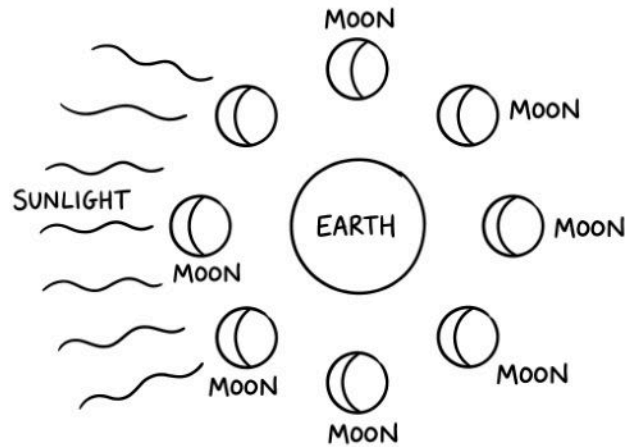
The Curious Kids Big Book of Why

Ages 6 to 12, 224 pages

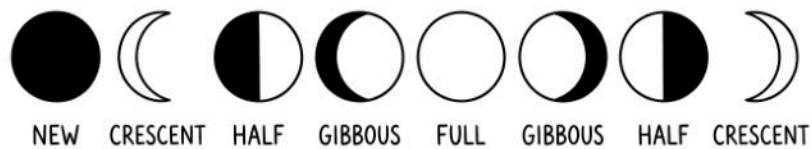


5 sample pages follow. Tap this page to see the whole book.

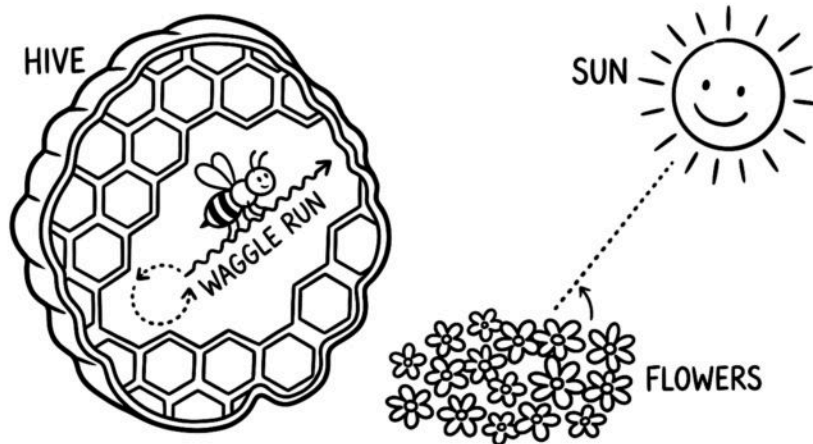
Why does the Moon look like it is changing its shape every night?



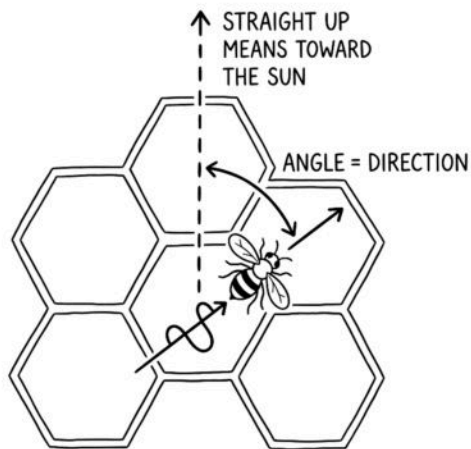
The Moon does not actually change its shape, and it does not make its own light. What we call moonlight is actually just sunlight reflecting off the Moon's surface like a giant mirror. As the Moon travels in a circle around Earth, the Sun always lights up exactly half of it. From our viewpoint on Earth, we see different amounts of that lit-up half depending on the Moon's position. These changing views are called the Moon's phases.



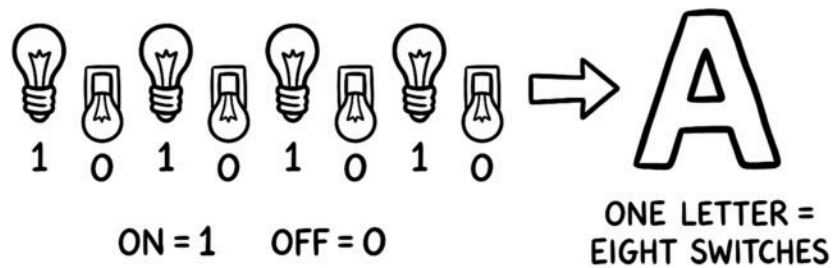
Why do honeybees do a waggle dance?



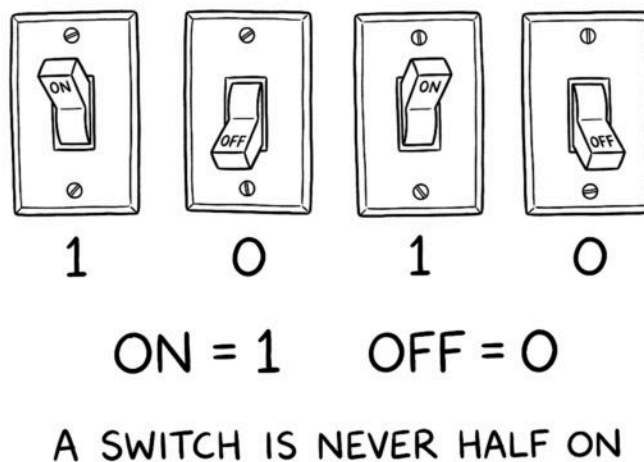
A honeybee that finds a good patch of flowers comes home and dances on the honeycomb. She runs in a straight line, wiggling her body, then loops back and repeats. Inside the dark hive she can feel which way is up. The angle of her straight run away from straight up is the same angle her sisters must fly away from the Sun. The longer she waggles, the farther the food is. Karl von Frisch worked this out and won a Nobel Prize in 1973.



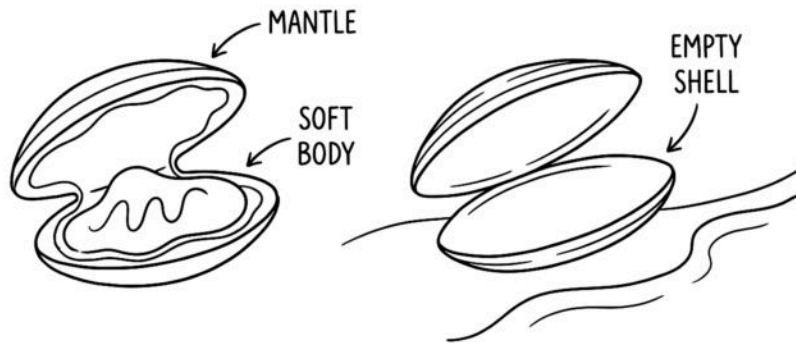
Why do computers use only ones and zeros?



Inside a computer are billions of tiny switches called transistors. A switch is easy to build and easy to read, because it is either on or off, with nothing in between. Even if the electricity wobbles a little, a computer can still tell on from off, so it almost never makes mistakes. We write on as 1 and off as 0. Any number, word, picture or song can be spelled out in those two symbols.



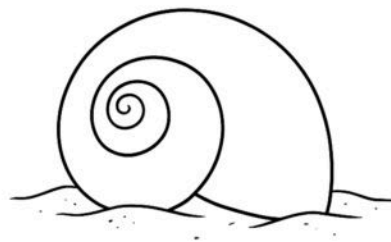
Why do seashells wash up empty?



A seashell is not a home someone moved out of. It is the skeleton of a soft animal called a mollusk, such as a snail, clam, or oyster. A body layer called the mantle builds the shell from calcium carbonate, the mineral in chalk. When the animal dies, or a crab or bird eats it, the soft body rots away or is swallowed. Only the hard shell is left, and waves roll it onto the beach.

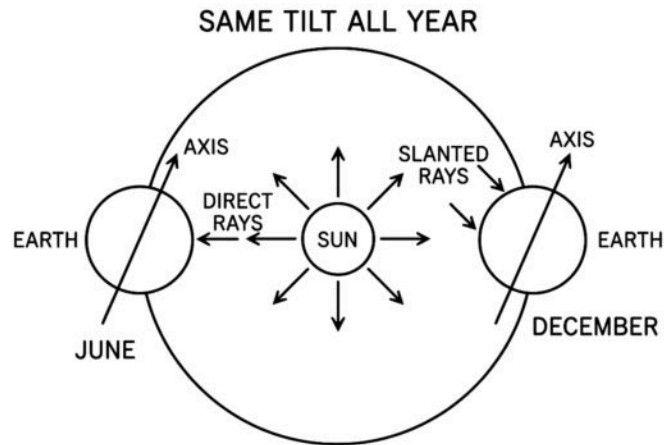


THE ANIMAL MAKES
THE SHELL

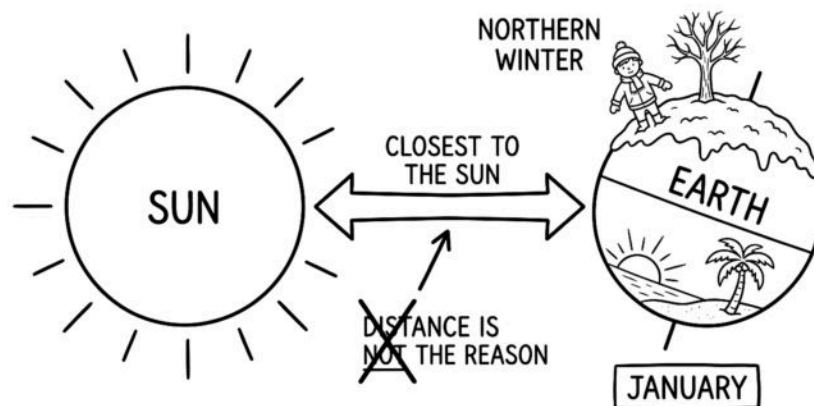


WHAT WASHES UP
IS THE HOUSE

Why do the seasons come at different times around the world?

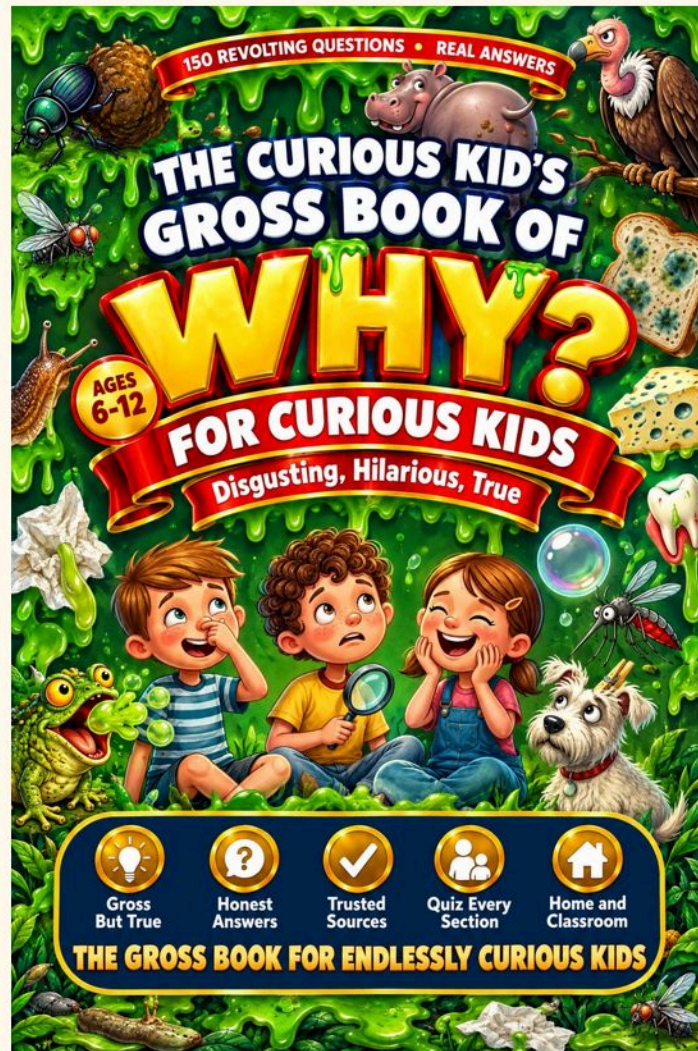


Earth spins on an axis that leans over by about 23.5 degrees, and that lean keeps pointing the same way all year. In June the northern half tips toward the Sun and catches its rays more directly, so the north has summer while the south has winter. Six months later the two swap over. Distance is not the reason. Earth is actually closest to the Sun in January, in the middle of northern winter.



The Gross Book of Why

Ages 6 to 12, 150 revolting questions



5 sample pages follow. Tap this page to see the whole book.

8 Why do dogs fart?

For much the same reasons people do. Every time a dog eats, it swallows a little air along with its food, and that air has to leave the body somewhere. Far more of a dog's gas comes from bacteria living in its large intestine, which break down bits of food the dog's own gut could not fully digest, releasing gas as they work. What a dog eats matters too: ingredients that are harder to digest give the bacteria more to ferment, and vets say hydrogen sulphide, one of the gases they make, is the main reason dog farts smell so bad. Scientists have even tested treats designed to make dog farts smell less strong, with some success.



BIG NUMBER

86

Percentage drop in really bad-smelling dog farts when scientists gave dogs treats containing charcoal, yucca and zinc, compared with a plain treat.

Journal of the American Veterinary Medical Association (2001), Administration of charcoal, Yucca schidigera, and zinc acetate to reduce malodorous flatulence in dogs

14 Why do we cough?

A cough is your body blasting something out of your airways: a bit of dust, a crumb that went the wrong way, or the extra mucus that comes with a cold. It happens in three quick stages. First you take a deep breath in. Then the opening at the top of your windpipe, called the glottis, shuts tight while the muscles of your chest, tummy and diaphragm squeeze hard, so the pressure inside your chest shoots up. Then the glottis flies open and the trapped air bursts out, at up to twelve litres a second, breaking loose mucus into droplets and making the coughing sound. Doctors say coughing is an important way of keeping your throat and airways clear.



TRY THIS

Put a hand flat on your tummy and give one good cough. Feel the muscles clench a moment before the sound? That squeeze, with the glottis shut, is what builds up the pressure for the blast.

Am J Respir Crit Care Med (2020), Coughs and Sneezes: Their Role in Transmission of Respiratory Viral Infections, Including SARS-CoV-2

2 Why is earwax different colours?

Earwax comes in two genetically determined types, and which one you have is written into your genes, just like eye colour. Dry earwax is brittle and flaky, and its colour ranges from pale to a brownish-grey. Wet earwax is darker, sticky and can look almost like honey. Scientists have traced this difference to a single gene, and the pattern follows simple rules: the version of the gene for wet earwax is dominant, and the version for dry earwax is recessive, so you need two copies of the dry version, one from each parent, to have the dry type. Neither type is better or worse, they are simply different, and families often share the same one.



BIG NUMBER

2

genetically determined types of earwax there are, dry and wet, and which one you have depends on a single gene passed down from your parents.

J Prim Care Community Health (2020), Cerumen Management

10 Why do cockroaches survive almost anything?

Cockroaches seem to survive almost anything because they combine tough bodies, flexible diets and slow cell division with rapid breeding, not because they are magic. They recycle decaying matter, taste carefully, and some can go weeks without food. Their cells divide mainly around moulting, so radiation damages them less than it damages constantly dividing human cells; tests show they can endure doses many times higher than a lethal human dose, though flour beetles beat them. They still die in the heat and blast of a nuclear fireball at ground zero. Their real edge for everyday life is surviving harsh homes, poisons and scarce meals better than most household insects.



BIG NUMBER

10,000

In radiation tests, cockroaches survived about 10,000 rad, roughly ten times a dose that would kill a human, yet flour beetles outlasted them at higher levels.

University of Florida IFAS Entomology

18 Why do sea cucumbers throw out their insides?

Soft sea cucumbers cannot dash away from crabs or fish, so some use a startling defence. When threatened, they may eject internal organs through the rear end, a process called evisceration. NOAA Fisheries notes California sea cucumbers can expel organs to deter predators and then regrow them. The University of Texas Marine Science Institute explains that the free snack distracts the attacker long enough for the cucumber to crawl to safety, and lost parts regenerate over weeks. Some species instead shoot sticky Cuvierian tubules that tangle foes. It looks extreme, yet it is a controlled escape plan, not a fatal accident.



TRY THIS

Gently squeeze a soft stress ball, then release it. Sea cucumbers can stiffen or soften their bodies on purpose; evisceration is another controlled body change.

University of Texas Marine Science Institute, Slippery Characters

Want more?

Scan for games, a why of the day,
and every book in the club.



ainjection.github.io/curious-kids